

Harry Rogers

Email: harry@harry-rogers.com

Hardware Test Engineer & Land IPT Test Point of Contact

Location: Sussex, United Kingdom

Specialising in Hardware-in-the-Loop (HIL) • Electro-Optics Testing • FPGA Verification

LinkedIn:

[linkedin.com/in/harryrogers073](https://www.linkedin.com/in/harryrogers073)

Portfolio: harry-rogers.com

WINNER

The IET Prize
2026

• Awarded by The Institution of
Engineering and Technology

First Class 80% BEng (Hons) Electronic &
Computer Engineering

• EXECUTIVE SUMMARY

Accomplished **Hardware Test Engineer & FPGA Verification Specialist** with a First-Class 80% BEng (Hons) in Electronic and Computer Engineering and recipient of **The IET Prize 2026**. Proven technical expertise spanning military-grade electro-optical surveillance system testing, hardware-in-the-loop (HIL) testbench design, and automated digital logic verification. Developer of **VAIDAR**, an automated Python-native execution engine bridging machine learning stimulus generation with physical AMD Xilinx Artix-7 FPGA silicon for accelerated coverage closure. Highly experienced in full-lifecycle Integration, Verification & Validation (IV&V), requirements traceability, protocol debugging, and cross-functional engineering collaboration.

• CORE TECHNICAL COMPETENCIES MATRIX

FPGA & Digital Logic Design

Verilog HDL, VHDL, AMD Xilinx Vivado Design Suite, Artix-7, Zynq-7000, RTL Simulation (ModelSim/Vivado Simulator), Logic Synthesis, XDC Timing Constraints, Static Timing Analysis (STA), Finite State Machines (FSM).

Hardware-in-the-Loop (HIL) & Automation

Python HIL Test Engines, PySerial, Automated Test Equipment (ATE), FAT/SAT/ATP Test Suites, Signal Injection Rigs, Real-Time Hardware Telemetry, Automated Discrepancy & Log Triage.

Electro-Optics & Systems Testing

Hawkeye Electro-Optical Surveillance Directors, Daylight HD Optics, Cooled Thermal Imagers, Eye-Safe Laser Rangefinders, Optical Boresighting, Environmental Qualification (ESS Thermal & Vibration), DOORS Traceability.

Protocols & Lab Instrumentation

Digital Storage Oscilloscopes, Logic Analysers, Spectrum Analysers, JTAG Boundary Scan, UART, PCIe, Ethernet, SPI, I2C, CAN Bus, RS-422 / RS-485, Camera Link, SDI, High-Speed Differential Signals.

• PROFESSIONAL WORK EXPERIENCE

Hardware Test Engineer & Land IPT Test Point of Contact

June 2026 – Present

Chess Dynamics • Horsham, West Sussex, UK

Full-Time Industry

Specialist defence enterprise manufacturing high-precision electro-optical tracking, surveillance directors, and stabilised payloads for allied defence programmes.

- Land IPT Test Point of Contact:** Serve as the primary test department point of contact for the Land Integrated Portfolio Team (IPT), advising project managers and systems leads on test scoping, verification feasibility, and qualification schedules.
- Factory Acceptance Testing (FAT):** Run rigorous Factory Acceptance Testing on Hawkeye electro-optical directors, verifying daylight HD cameras, cooled continuous-zoom thermal imagers, and laser rangefinders against strict customer specifications.
- Hardware Diagnostics & Fault Isolation:** Conduct board-level fault diagnosis across complex multi-layer PCBs, slip-ring comms (RS-422, CAN, Ethernet), motor controller feedback loops, and video processing pipelines.
- Optical Boresighting & Calibration:** Perform precision optical boresighting and laser rangefinder alignment to achieve high-accuracy weapon system and tracking alignment.
- Environmental Qualification (ESS):** Coordinate environmental stress screening, including thermal chamber cycling (-40°C to +70°C) and multi-axis vibration testing to military standards.
- Documentation & Compliance:** Author formal Factory Acceptance Test reports, engineering discrepancy logs, and Standard Operating Procedures (SOPs).

Systems Engineer — Integration, Verification & Validation (IV&V) Placement

June 2024 – June

2025

BAE Systems • Rochester, Kent, UK

12-Month Industrial Placement

Global tier-one defence and aerospace enterprise delivering mission-critical electronic systems and active vehicle electronics.

- **IV&V Lifecycle Execution:** Executed end-to-end Integration, Verification & Validation (IV&V) procedures on safety-critical electronic systems hardware and testbench architectures.
- **Python Testbench Automation:** Developed modular Python-based automated test scripts for lab testbenches and signal injection hardware, significantly accelerating regression testing throughput.
- **Requirements Traceability (IBM DOORS):** Managed bidirectional requirements verification matrices in IBM DOORS, mapping high-level system requirements to granular hardware test cases with 100% audit traceability.
- **Signal Integrity & Simulation:** Verified electronic hardware subsystems using signal injection, high-bandwidth oscilloscopes, and MATLAB/Simulink models, including Frequency Response Analysis (FRA).
- **Anomaly Debugging & ERB Reviews:** Collaborated with multidisciplinary hardware, software, and systems teams to diagnose hardware root-cause anomalies, authoring defect reports presented at Engineering Review Boards.
- **STEM Outreach:** Delivered interactive "STEM in a Box" engineering workshops to over 400 school students across Kent as a certified STEM Ambassador.

Co-President, Vice President & Web Architect

October 2025 – June 2026

Brighton Real Ale Society • University of Brighton, UK

- Elected executive leader managing society administration, event licensing, risk assessments, and financial budgeting for 100+ active university members.
- Architected, developed, and deployed the society's official web application ([bras-website](#)) using Next.js (React 19), Prisma ORM, and Supabase PostgreSQL with automated CI/CD on Vercel.

• FEATURED ENGINEERING & RESEARCH PROJECTS

VAIDAR: AI-Driven Hardware-in-the-Loop FPGA Verification Engine

AMD Xilinx Artix-7

Final Year Capstone Project • Grade: First Class 91% (A+) • The IET Prize Winner 2026

Architected an automated hardware-in-the-loop verification engine bridging host software directly to physical FPGA silicon over high-speed 921,600-baud UART to solve the 70% lifecycle bottleneck of software simulation.

Physical Hardware Testbed

High-speed serial transport between host Python runtime and synthesised 16-bit ALU on Digilent Nexys A7 FPGA silicon.

Machine Learning Stimulus Synthesis

Integrated adaptive stimulus generation and automated LLM discrepancy triage, delivering 99.9%+ logic coverage closure.

- Achieved **968.9 tests/second** throughput on physical silicon (~44,000x faster than manual bench testing with 100% physical accuracy).

Robot Wars Autonomous Combat Platform & Motor Drives

2nd Place Championship • Grade: First Class 82%

Engineered high-current MOSFET dual H-bridge motor drives, SolidWorks CAD aluminium chassis, and PIC microcontroller firmware for autonomous combat buggy.

VHDL Multi-Floor Elevator Controller (Intel DEO-Nano FPGA)

Level 5 Digital Systems Design • Grade: First Class 84%

Implemented robust finite state machines (FSM), debounced input registers, and hardware interlocks in IEEE VHDL, verified via ModelSim RTL testbenches.

• EDUCATION & ACADEMIC DISTINCTION

BEng (Hons) Electronic & Computer Engineering
University of Brighton • First Class 80%

Graduated July 2026

Level 6 (Final Year) — 80%

- Individual Project (VAIDAR): 91%
- Communications & Networks: 90%
- Embedded Systems 3 (MATLAB): 84%
- Digital Signal Processing: 68%

Level 5 (Second Year) — 77%

- Digital Systems (FPGA VHDL): 84%
- Engineering Design (Robotics): 82%
- Embedded Systems 2 (XC8 C): 79%
- Electrical Engineering 2 (Power): 76%

Level 4 (First Year) — 78%

- Analogue & Digital 1: 87% (IET Prize)
- Intro Electrical Engineering: 87%
- Engineering Mathematics: 84%
- Embedded Systems 1 (MPASM): 79%

• OFFICIAL ENGINEERING HONOURS & DISTINCTIONS

WINNER The IET Prize 2026

Awarded by **The Institution of Engineering and Technology (IET)** for outstanding academic distinction and graduating First Class 80% in BEng Electronic & Computer Engineering at the University of Brighton.

2026

Signatories: Ed Almond (Chief Executive), Dawn Ohlson (IET President 2025–26), Warren East (Board of Trustees).

WINNER IET Sussex Prize 2023

Awarded by **The Institution of Engineering and Technology (IET)** for outstanding first-year academic performance across Analogue & Digital Electronics and Electrical Engineering.

2023

• ADDITIONAL WORK HISTORY & COMMERCIAL LEADERSHIP

Customer Service Assistant & Team Member
bp / M&S Simply Food • Sussex, UK
Forecourt safety compliance, hazardous materials handling, fresh food auditing, and cash handling procedures.

Dec 2023 – June 2026

Shift Manager
McDonald's • Eastleigh, Hampshire, UK
Directed high-tempo shifts with up to 30 personnel, labour scheduling, hygiene audits, safe drops, and customer conflict resolution.

Aug 2020 – Mar 2024

Bicycle Courier
Deliveroo • Brighton, UK
Time-critical urban food logistics across complex city topography under strict delivery time windows.

Sept 2023 – June 2024

Volunteer Computer Refurbisher
Jamie's Computers • Southampton, UK
Hardware diagnostics, component testing (RAM/PSU/drives), secure data wiping, and OS installs for community reuse.

May 2021 – Oct 2021

• TECHNICAL TOOLING & SYSTEMS INDEX

HDL & EDA

- Verilog HDL
- IEEE VHDL
- AMD Xilinx Vivado
- ModelSim RTL
- Quartus Prime
- XDC Constraints

Software & Scripting

- Python 3.x / PySerial
- Embedded C / C++
- MATLAB / Simulink
- DXL (IBM DOORS)
- TypeScript / React
- Linux / Git / CI/CD

Bus Protocols

- RS-422 / RS-485
- CAN Bus
- Ethernet / TCP/IP
- UART / SPI / I2C
- MIL-STD-1553
- Camera Link / SDI

Lab Instrumentation

- Storage Oscilloscopes
- Logic Analysers
- Spectrum Analysers
- Thermal Blackbodies
- Optical Collimators
- JTAG Boundary Scan